Jam Kabeer Ali Khan

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RESEARCH INTERESTS

I work on the theory and implementation of programming languages and verification, building tools with *foundational* guarantees. My research spans type systems, program logics, Horn solving, refinement types, and quantum verification.

EDUCATION

Sept 2021 – June 2025 **University of Hong Kong** Hong Kong
Major in Computer Science & Minor in Mathematics
Fully-Funded Tuition Scholarship

Jan 2024 – May 2024 **Northeastern University** Boston, USA
Semester Exchange (Computer Science & Mathematics)

RESEARCH EXPERIENCE

University of California, San Diego Jan 2026 – Present
Research Collaboration Remote
Supervisor: Ranjit Jhala · Collaborator: Nico Lehmann

- Design of Zap, a *proof-producing* CHC algorithm based on NNF unfolding.
- Lean4 metaprogramming for proof-producing CHC-solving tactics.
- Verification of CHC-based imperative and functional verifiers.
- SMTLib-to-Lean: tooling to translate SMT queries into well-typed Lean4 propositions as an escape hatch.
- (Ongoing) Liquid types for automated functional verification of Lean4 code.

Max Planck Institute for Security & Privacy Apr 2025 – Sept 2025
Research Intern Germany
Supervisor: Gilles Barthe · Collaborator: Michael Walter

- Compiler correctness for probabilistic programs compiled to quantum programs.
- Input-dependent and worst-case cost analysis of probabilistic programs via quantization.
- Implementation of quantum algorithms, including Simon's and Grover's, as Traq primitives in Haskell.
- Verification case studies using sound and complete logic for quantum-classical programs.
- Implementation of CQOTL, an interactive prover in OCaml for relational reasoning about quantum programs.
- Lean4 mechanization of quantum computing infrastructure, including Dirac and orthonormal bases, as a backend for CQOTL.

HKU Programming Languages Group Nov 2025 – Present; Aug 2024 – Apr 2025
Research Hong Kong
Supervisor: Bruno C. d. S. Oliveira

- Type-directed elaboration of a module language to *first-class* environments.
- Lean4 mechanization of separate compilation, semantics preservation, and type preservation.
- Source language design for the polymorphic calculus FE.
- Language prototypes in Haskell and OCaml.

HKU Quantum Lab June 2024 – Aug 2024
Research Intern Hong Kong
Supervisor: Ravishankar Ramanathan

- Design of DI-QKD protocol based on simplex measurements and finding guessing probability curves.
- Used semidefinite programming (NPA Hierarchy) to find SOS decomposition to show self-testing for Bell Inequality.

INDUSTRIAL EXPERIENCE

Standard Chartered - Core Strats Oct 2025 – Present
Functional Programmer (Haskell), full-time Hong Kong

- Profiling, optimisation and maintenance of the Shake build system in Haskell.
- Implementation of a gRPC runtime in C++ for Mu, a strict Haskell variant.
- Protoc compiler extensions to generate Mu code from protobuf files.
- Automated tool to analyse dependencies and identify module closures, unused exports and functions.

PUBLICATIONS

Peer-reviewed papers

LICS 2026 **Complete Relational Logic for Infinite-Dimensional Quantum Programs with Unbounded Assertions** [PDF]
Gilles Barthe, Minbo Gao, **Jam Kabeer Ali Khan**[*], Matthijs Muis, Ivan Renison, Keiya Sakabe, Michael Walter, Yingte Xu, Tianshi Yu and Li Zhou

PLDI 2025 **Capabilities as First-Class Modules with Separate Compilation** [PDF]
SRC **Jam Kabeer Ali Khan**
Gold Award (Undergraduate Category)

Manuscripts

Submitted **Linking as Evaluation: Separate Compilation via First-Class Environments** [PDF]
OOPSLA'27 **Jam Kabeer Ali Khan**, Jinhao Tan and Bruno C. d. S. Oliveira

Preprint **Foundational Constraint Solving for Expressive Refinement Typing** [PDF]
arXiv **Jam Kabeer Ali Khan**, Petros Markopoulos, Nico Lehmann and Ranjit Jhala

Draft **Polymorphic First-Class Environments** [PDF]
Jinhao Tan, **Jam Kabeer Ali Khan** and Bruno C. d. S. Oliveira

Preprint **Traq: Estimating the Quantum Cost of Classical Programs** [PDF]
arXiv Anurudh Pedhuri, **Jam Kabeer Ali Khan**, Gilles Barthe and Michael Walter

[*] Authors listed alphabetically.

PRESENTATIONS & TALKS

HKU PL Seminar **Flex: *Proof-producing* CHC solver** [Slides]

PLanQC '26 **Traq: Estimating the Quantum Cost of Classical Programs** [Slides] [Extended Abstract] [Talk]

PLDI '25 SRC **Capabilities as First-Class Modules with Separate Compilation** [Talk] [Poster]

HKU Seminar **Maximal Violation of a Measurement Protocol with Simplex Bell Inequality** [Slides]

SELECTED SOFTWARE

Lean4 **Flex: Foundational Constraint Solver** [GitHub]
Certified *Constraint Horn Clause* (CHC) solving to enable expressive refinement typing.

Lean4 **SMTLib-to-Lean4: Tool to send SMT queries to Lean4** [GitHub]
Translation of SMT Queries to Lean4 Proposition, including CHCs.

Haskell **EnvML: Polymorphic *First-Class* Environments** [Playground] [GitHub]
An ML-like language with dynamic module composition elaborated to a variant of System F with *first-class* environments.

OCaml, Lean4 **SCE: Language and calculus** [Playground] [GitHub] [Mechanization]
A mechanized calculus with *first-class* linking and separate compilation; Compiler based on SCE calculus and WASM backend.

TEACHING EXPERIENCE

Fall 2024 **Object-Oriented Programming in Java (COMP2396)**
Fall 2023 *Teaching Assistant*, University of Hong Kong
Instructor: Kenneth Wong

Spring 2023 **Computer Programming in C++ (ENGG1340)**
Teaching Assistant, University of Hong Kong
Instructor: Dr. Tat Wing Chim (T. W. Chim)

Feb 2022 – Aug 2023 **Mathematics & Computer Science**
Tutor, All Round Education Academy — Hong Kong

HONORS & AWARDS

2025 PLDI'25 SRC Gold Award
2024 Rosita King Ho Scholarship
2022 – 2024 HKU Undergraduate Entrance Scholarship for Outstanding Academic Talents
2021 – 2024 HKSAR Government Scholarship Fund
2023 Cathay Hackathon: 2nd runner-up (Free flight to Boston)
2021 HKU Foundation Entrance Scholarship
2020 Top in Sindh and Balochistan (GCE A-levels CS & Physics)
2018–20 100% merit scholarship to attend GCE A-levels at Nixor College.

SERVICE

POPL'27 Artifact Evaluation Committee
APLAS'26 Student Volunteer
ICFP'26 Artifact Evaluation Committee
CAV'25 Student Volunteer

OTHER EXPERIENCE

All Round Education Academy — Mar 2023 – Nov 2023
Software Developer (Freelance) — Hong Kong

ActusRayPartners Limited Hedge Fund — June 2023 – Aug 2023
Technology Intern — Hong Kong

HKU Sleep Lab — Apr 2022 – Oct 2022
Student Research Assistant — Hong Kong

SKILLS

Languages **Haskell**, **OCaml**, C++, Racket, Java, Python
Proof Assistants Rocq, Lean4
Tools Git & LaTeX

REFERENCES

Gilles Barthe
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Scientific Director
Max Planck Institute for Security and Privacy
Bochum, Germany
Google Scholar

Ranjit Jhala
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